

DS 404: Information Security and Privacy

Course Code	DS 404
Course Name	Information Security and Privacy
L-T-P-C	3-0-0-3
Prerequisites	DS 203: Mathematical Foundations of Data Science I
Intended for	B.Tech.
Distribution	Discipline core for B. Tech. Data Science and Engineering, Discipline elective for B. Tech. Computer Science and Engineering, Electrical Engineering, Free elective for other B. Tech. disciplines
Semester	Even/Odd.
Approval	29th BoA (Mar 2019); 20th Senate (Apr 2019), in the B.Tech Data Science and Engineering curriculum
Revised	65th BoA (Aug 2026) - credits changed to 3

Course Objectives

During the last two decades, there has been a major paradigm shift in generation, processing, communication, and storage of information from predominantly analog domain to digital domain for the reasons such as ease of implementation, better efficiency, greater robustness against noise, and enhanced performance and security. This shift has not only resulted in opening up exciting new possibilities in numerous areas of sciences, engineering, and humanities, but have also created plethora of possibilities for the misuse of this data, raising serious security, privacy and ethical challenges. To address these challenges, DSE curriculum must not only introduce the students to spectrum of such challenges, but also equip them with various concepts, tools, techniques. This will enable the students to better understand information security, privacy, and ethical challenges and appreciate the current developments at the frontiers of these areas, both in theory and application, and prepare them to contribute to further advancement of such frontiers.

On completion of this course, students should be able to demonstrate their understanding of and apply methods of information security, privacy, and ethics to subsequent courses in data mining, pattern recognition, knowledge representation, information systems, to name a few. In particular, students should be able to

- demonstrate understanding and awareness of information security, privacy, and ethical issues in a given application scenario.
- propose practical solutions to address such issues using relevant concepts, tools, and techniques

Course Modules

- **Introduction to information security:** Information security models; attacks, threats, vulnerabilities, and risks. Operations security: Haas' Laws. Identification and authentication: identity verification, falsifying identification, multifactor and mutual authentication, passwords, biometrics, hardware tokens, performance evaluation. Authorization and access control: principle of least privilege, access control lists, and access control methodologies, physical security and access controls. Auditing and accountability: non-repudiation, deterrence, intrusion detection and prevention, logging, monitoring, assessments. (8 lectures)
- **Cryptography:** Protocols (key exchange, public key cryptography, secret sharing), techniques (key length, key management, etc), cryptographic algorithms (mathematical background, data encryption standard, block and stream ciphers, public-key, digital signatures). (8 lectures)
- **Network security:** Protecting networks and network traffic, mobile device security, network security tools. (6 lectures)
- **Operating system security:** OS hardening, protecting against malware, firewalls and host intrusion detection, OS security tools. (4 lectures)
- **Application security:** Software development vulnerabilities, web security, database security, and application security tools. (6 lectures)
- **Information privacy:** Static and dynamic data anonymization and threats to anonymization, privacy in synthetic and test data, privacy regulations. (6 lectures)
- **Information Ethics:** Ownership, privacy, anonymity, validity, algorithmic fairness, societal consequences, code of ethics, attributions. (4 lectures)

Text Books

1. Andress, J. and Winterfeld, S., *The Basics of Information Security*, 2/e, Syngress, 2014.
2. Venkataraman, N. and Shriram, A., *Data Privacy: Principles and Practice*, Chapman and Hall/CRC, 2016.

References

1. Guise, P. D., *Data Protection*, Routledge, 2017.
2. Katz, J. and Lindell, Y., *Introduction to Modern Cryptography*, Chapman and Hall/CRC, 2015.
3. Torra, V., *Data Privacy: Foundations and the Big data Challenge*, Springer, 2017.
4. Weigand, A., *Data for the People*, Basic Books, 2017.